AGA KHAN UNIVERSITY EXAMINATION BOARD

SECONDARY SCHOOL CERTIFICATE

CLASS IX EXAMINATION

MAY 2015

Chemistry Paper II

Time: 2 hours 25 minutes Marks: 40

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are EIGHT questions. Answer ALL questions. Questions 7 & 8 each offer TWO choices. Attempt any ONE choice from each.

3. When answering the questions:

   Read each question carefully.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a simple calculator if you wish.

Q.1. (Total 3 Marks)

$$___ C_2H_{4(g)} + ___ O_{2(g)} \rightarrow ___ CO_{2(g)} + ___ H_2O_{(l)}$$

a. Balance the above equation using the simplest possible whole numbers. (1 Mark)

b. Identify the type of chemical reaction taking place in the above equation. (1 Mark)

_____

_____

c. How many moles of oxygen gas (O_2) are required to react with 3.5 moles of ethene gas (C_2H_4)?

[NOTE: Use the above balanced chemical equation for the calculation.] (1 Mark)

_____

_____

Q.2. (Total 4 Marks)

The table below shows the number of protons, electrons and neutrons for substances A, B, C and D.

Substance	Number of Protons	Number of Electrons	Number of Neutrons
A	17	18	18
B	15	15	16
C	11	10	12
D	6	6	7

Based on the given data, answer the following questions.

a. Identify the substances A to D according to the table below. (3 Marks)

Positive Ions	Negative Ions	Neutral Atoms

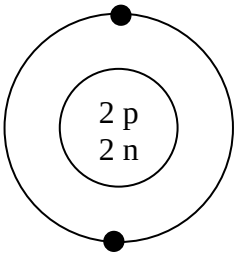
b. Give the letter of the substance which is an isotope of $^{12}_{6}C$. (1 Mark)

_____

_____

Q.3. (Total 5 Marks)

Consider the given structure of an atom of an element X to answer the following questions.



a. Identify the element X. (1 Mark)

_____

_____

b. Name another element belonging to the same group as element X. (1 Mark)

_____

_____

c. State any THREE properties that all the elements of this group have in common. (3 Marks)

_____

_____

_____

_____

_____

_____

Q.4. (Total 4 Marks)

Diamond and graphite are allotropes of carbon.



Diamond

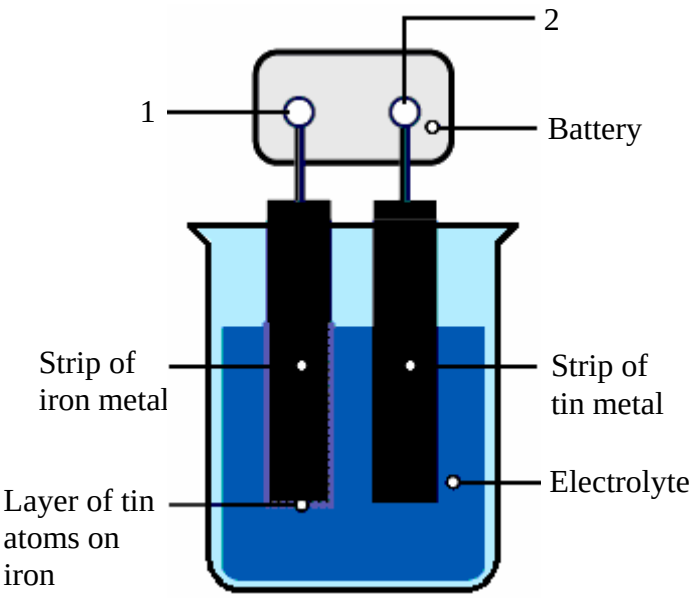


Graphite

Distinguish between these two forms of carbon on the basis of their electrical conductivity. Also give an explanation for the difference.

Q.5. (Total 5 Marks)

a. In the electrolytic cell shown below, an iron strip is being plated with tin.



i. Identify the terminals 1 and 2 of the battery as positive or negative. (1 Mark)

Terminal 1 is _____

Terminal 2 is _____

ii. Suggest a suitable electrolyte for the above electroplating process. (1 Mark)

_____

_____

iii. Give a reason why an electrolyte is required for this process. (1 Mark)

_____

_____

b. In the given redox reaction, zinc can displace copper because zinc is more reactive than copper.

$$Zn_{(s)} + Cu(NO_3)_{2(aq)} \rightarrow Zn(NO_3)_{2(aq)} + Cu_{(s)}$$

Identify which substance acts as an oxidizing agent and which one acts as a reducing agent in the given reaction. (2 Marks)

Oxidizing Agent: _____

Reducing Agent: _____

Q.6. (Total 4 Marks)

a. Sodium ($_{11}$Na) and magnesium ($_{12}$Mg) are both metals. They react vigorously with chlorine to form ionic salts as shown in the following chemical reactions.

Sodium with chlorine: $2Na_{(s)} + Cl_{2(g)} \rightarrow 2NaCl_{(s)}$

Magnesium with chlorine: $Mg_{(s)} + Cl_{2(g)} \rightarrow MgCl_{2(s)}$

Describe why there is a difference between the formulae of both the ionic salts, i.e. NaCl and $MgCl_2$. (2 Marks)

b. Give TWO reasons why alkali metals are not found in a free state in nature. (2 Marks)

Q.7. (Total 8 Marks)

EITHER

a. A group of students classified four different substances into compounds or mixtures as follows.

Compound	Mixture
Distilled water	Soil
Carbon dioxide gas	Sea water

With reference to distilled water, carbon dioxide gas, soil and sea water, state FOUR characteristics of compounds and FOUR characteristics of mixtures that helped the students in their classification.

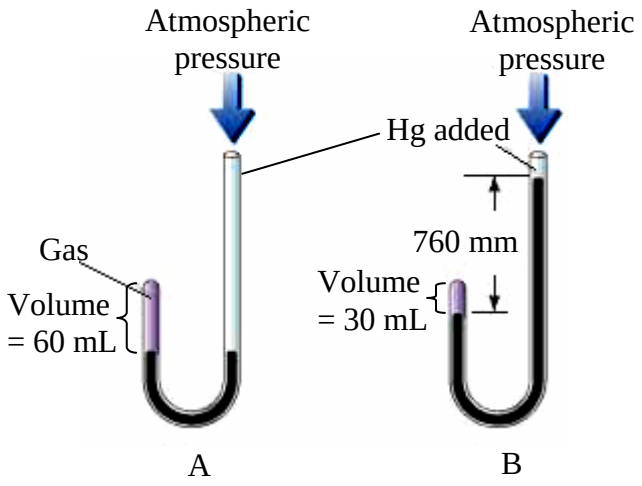
OR

b. Differentiate between polar and non-polar covalent bonds on the basis of electronegativity. Also describe the formation of a polar covalent bond in a hydrogen fluoride (HF) molecule on the basis of their valence shell electronic configuration. You may use diagram if you wish.

Q.8. (Total 7 Marks)

EITHER

a. Robert Boyle (1627-1691), an Irish chemist, performed the first quantitative experiment on gases using a J-shaped tube to study the relationship between the pressure of the trapped gas and its volume. He added mercury to the tube to study the effect of a change in pressure on the volume of the gas.



State Boyle's law. Give a mathematical expression and sketch a graph to represent Boyle's law. Use Boyle's law to explain the differences in the volume of gas observed in tube A and B above.

OR

b. Define solubility. With the help of examples, describe the effect of temperature and pressure on solubility of solids and gases.

END OF PAPER

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Please use this page for rough work